1. **Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.**

Code:

```
import java.util.ArrayList;
import java.util.Scanner;

public class TodoList {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // ArrayList to store tasks
        ArrayList<String> taskList = new ArrayList<String>();

        int choice;

        do {
            System.out.println("\n----- TO DO LIST -----");
            System.out.println("1. Add Task");
            System.out.println("2. Remove Task");
            System.out.println("3. View Tasks");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");

            choice = sc.nextInt();
            sc.nextLine();

            if (choice == 1) {
                // Add task
                System.out.print("Enter task: ");
```

```java
        String task = sc.nextLine();
        taskList.add(task);
        System.out.println("Task Added.");

    } else if (choice == 2) {
        // Remove task
        if (taskList.size() == 0) {
            System.out.println("No tasks available.");
        } else {
            System.out.print("Enter task number: ");
            int n = sc.nextInt();

            if (n >= 1 && n <= taskList.size()) {
                taskList.remove(n - 1);
                System.out.println("Task Removed.");
            } else {
                System.out.println("Invalid task number.");
            }
        }

    } else if (choice == 3) {
        // Display tasks using StringBuffer
        if (taskList.size() == 0) {
            System.out.println("Task list is empty.");
        } else {
            StringBuffer sb = new StringBuffer();

            sb.append("\nTasks:\n");

            for (int i = 0; i < taskList.size(); i++) {
                sb.append((i + 1) + ". " + taskList.get(i) + "\n");
            }

            System.out.println(sb);
        }
```

```java
            } else if (choice == 4) {
                System.out.println("Program Ended.");

            } else {
                System.out.println("Invalid Choice.");
            }

        } while (choice != 4);

        sc.close();
    }
}
```

Output:

```
----- TO DO LIST -----
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter choice: 1

Enter task: Complete Assignment
Task Added.

----- TO DO LIST -----
Enter choice: 1

Enter task: Buy Books
Task Added.

----- TO DO LIST -----
Enter choice: 3

Tasks:
1. Complete Assignment
2. Buy Books

----- TO DO LIST -----
Enter choice: 2

Enter task number: 1
Task Removed.

----- TO DO LIST -----
Enter choice: 3

Tasks:
1. Buy Books

----- TO DO LIST -----
Enter choice: 4

Program Ended.
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Code:

```
import java.util.ArrayList;
import java.util.Scanner;

public class StudentCourse {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ArrayList<String> courses = new ArrayList<String>();

        int choice = 0;

        while (choice != 4) {

            System.out.println("\nCourse Registration");
            System.out.println("1. Add Course");
            System.out.println("2. Remove Course");
            System.out.println("3. View Courses");
            System.out.println("4. Exit");
            System.out.print("Enter Choice: ");
            choice = sc.nextInt();
            sc.nextLine();

            if (choice == 1) {
```

```
    System.out.print("Enter Course Name: ");
    String course = sc.nextLine();
    courses.add(course);
    System.out.println("Course Added");

} else if (choice == 2) {

    System.out.print("Enter Course Name to Remove: ");
    String course = sc.nextLine();

    if (courses.remove(course)) {
        System.out.println("Course Removed");
    } else {
        System.out.println("Course Not Found");
    }

} else if (choice == 3) {

    StringBuffer sb = new StringBuffer();
    sb.append("Registered Courses:\n");

    for (int i = 0; i < courses.size(); i++) {
        sb.append(courses.get(i)).append("\n");
    }

    System.out.println(sb);

} else if (choice == 4) {

    System.out.println("Program Ended");

} else {

    System.out.println("Invalid Choice");
}
```

```
        }

        sc.close();
    }
}
```

Output:

```
Course Registration
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 1

Enter Course Name: Java
Course Added

Course Registration
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 1

Enter Course Name: Python
Course Added

Course Registration
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 3

Registered Courses:
Java
Python

Course Registration
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 2

Enter Course Name to Remove: Java
Course Removed

Course Registration
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 3

Registered Courses:
Python
```

```
Course Registration
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice: 4

Program Ended
```